

In the following assume $\mathcal{U}$ is a monster model for T . One can identify $S_x(M)$ with $\mathcal{U}/\sim$ where $a \sim b$ iff $tp(a/M) = tp(b/M)$. Let $f : \mathcal{U} \rightarrow S_x(M), a \mapsto tp(a/M)$.

Esercizio 1. Let $A \subseteq S_x(M)$ open. Then tfae:

1. A p -open

2. for all $\phi(x, b)$ such that $\phi(\mathcal{U}, b) \subseteq A$ there exists $\theta(z) \in tp(b/\emptyset)$ such that for all $b' \in \theta(M)$ $\phi(\mathcal{U}, b') \subseteq A$

Proof. • 1) $\Rightarrow$ 2): Let $p(x) \in A$. Then there must exist $\psi_p(x, z), \theta_p(z) \in L$ such that $p(x) \in \neg R_{\psi_p, \theta_p}(S_x(M)) \subseteq A$. This is equivalent to saying that:

- $\exists b' \in \theta(M) \psi_p(x, b') \in p(x)$
- $\bigcup_{b' \in \theta(M)} \psi_p(S_x(M), b') \subseteq A$, where $\psi_p(q(x), b')$ holds iff $\psi_p(x, b') \in q(x)$. If we take $q(x) = tp(a/M)$, then saying $\psi(q(x), b')$ holds is the same as saying $\psi(a, b')$ holds. So if we step back from $S_x(M)$ to $\mathcal{U}$ the condition becomes $\bigcup_{b' \in \theta(M)} \psi_p(\mathcal{U}, b') \subseteq f^{-1}(A)$.

The first condition tells us that $p(\mathcal{U}) \subseteq \psi_p(\mathcal{U}, b')$. Then

$$\phi(\mathcal{U}, b) \subseteq \bigcup_{p(x) \models \phi(x, b)} p(\mathcal{U}) \subseteq \bigcup_{p(x) \models \phi(x, b)} \bigcup_{b' \in \theta_p(M)} \psi_p(\mathcal{U}, b')$$

By compactness we can extract

$$\phi(\mathcal{U}, b) \subseteq \bigcup_{i=1, \dots, n} \psi_i(\mathcal{U}, b'_i) \text{ with } b'_1 \in \theta_1(M), \dots, b'_n \in \theta_n(M)$$

This implies that $\mathcal{U} \models \forall x \left(\phi(x, b) \rightarrow \bigvee_{i=1, \dots, n} \psi_i(x, b'_i) \right)$. By elementarity M models the same. Now consider

$$\theta(z) : \forall x \left(\phi(x, z) \rightarrow \bigvee_{i=1, \dots, n} \exists y (\theta_i(y) \wedge \psi_i(x, y)) \right)$$

It is immediate that $\theta(z) \in tp(b/\emptyset)$. Now let $M \models \theta(c)$. Let $d_1, \dots, d_n \in M$ such that $M \models \forall x \left(\phi(x, c) \rightarrow \bigvee_{i=1, \dots, n} \theta_i(d_i) \wedge \psi_i(x, d_i) \right)$. By elementarity $\mathcal{U} \models \forall x \left(\phi(x, c) \rightarrow \bigvee_{i=1, \dots, n} \theta_i(d_i) \wedge \psi_i(x, d_i) \right)$. This is the same as $\phi(\mathcal{U}, c) \subseteq \bigcup_{i=1, \dots, n} \psi_i(\mathcal{U}, d_i) \subseteq A$. So we have $\theta(z)$ is the formula we were looking for.

• 2) $\Rightarrow$ 1): Since A open we can write

$$A = \bigcup_{\phi(\mathcal{U}, b) \subseteq A, b \in M} \phi(\mathcal{U}, b) \subseteq \bigcup_{\phi(\mathcal{U}, b) \subseteq A, b \in M} \neg R_{\phi, \theta_\phi}$$

where $\theta_\phi(z)$ is the formula given by hypothesis.

□